

TIC 1003150980

THIN MARGIN

Eclipsing binary candidate dossier - Star Vetter

BLS EPHEMERIS

Period 4.776509 d
Epoch (t0) 1628.1277 BTJD
Duration 0.0840 d
Depth 2.711 % (27,108 ppm)
SNR 176.6

CATALOG CROSS-MATCH (novelty)

VSX no match
ASAS-SN no match
Gaia eclipsing-binary table no match
Gaia variable classification no match
TESS EB catalog (Prsa+2022) no match
ExoFOP TOI list no match

INDEPENDENT CHECKS (f3i gates)

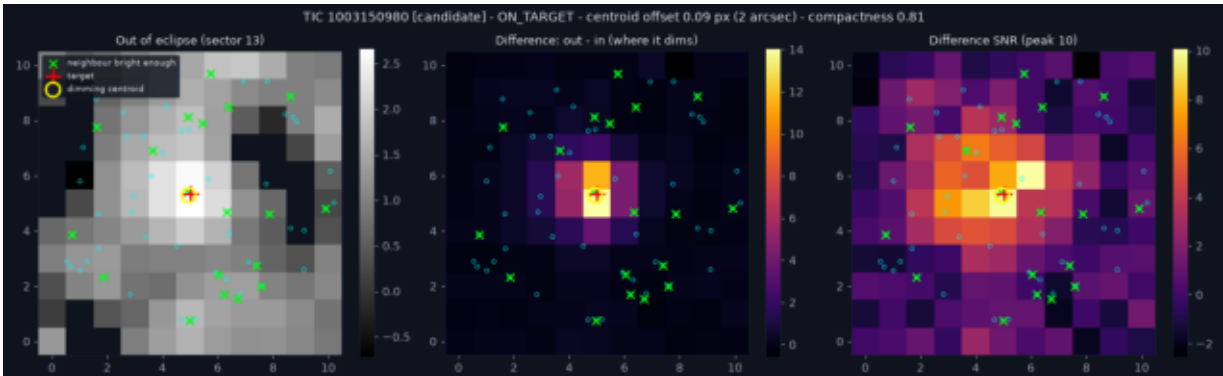
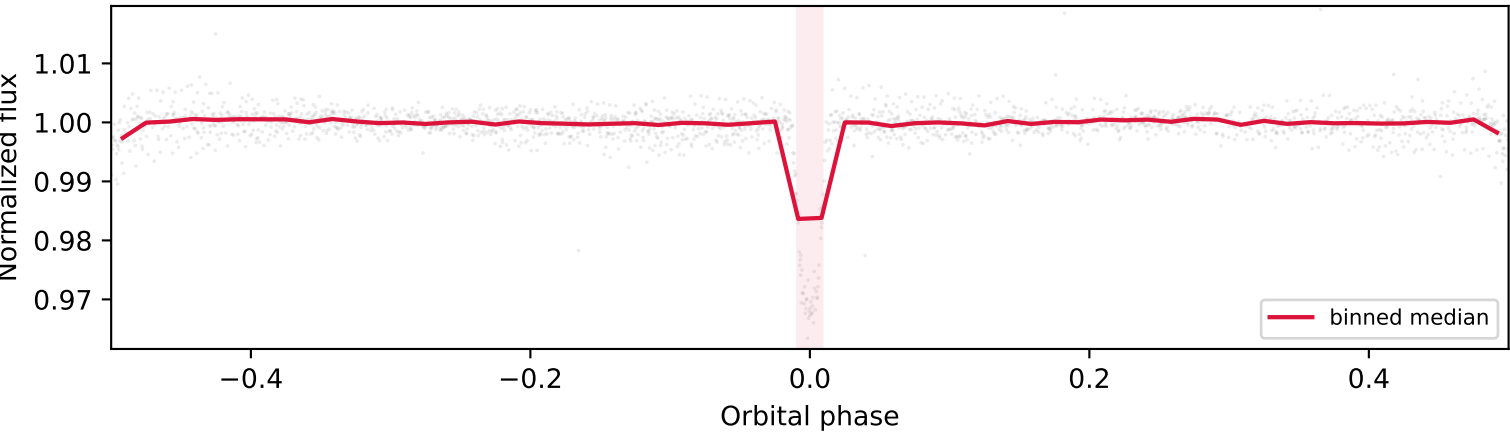
Eclipses observed 10 (gate >= 3)
Robust depth ratio 1.07 (gate >= 0.50)
Robust depth sigma 91.0 (gate >= 7.0)
Odd/even z-score 0.50 (consistent)
Secondary eclipse detected, phase -0.49, 5.3 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 13 light-curve dip in this sector: 87.0 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 10.08 (gate >= 5.0) Compactness 0.81 (gate >= 0.50)
Dimming centroid offset 0.09 px (2 arcsec) (gate <= 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 383652989
0.11 px away, dTmag -1.71 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.019 px (dimming centroid 0.089 px from target vs 0.108 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.